

COSC 594

DM Composition (III)

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Outline

- RECAP
- REGULARIZATION-BASED APPROACH
- CONSTRAINED APPROACH

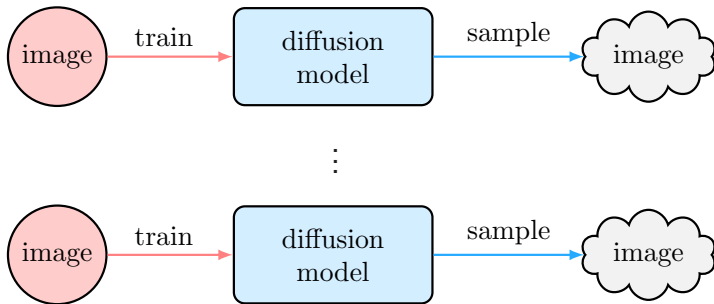
Outline

■ RECAP

Combination of DMs



⋮



$$x_0 \sim p^1(\cdot), \dots, p^K(\cdot)$$

How do we combine several pretrained DMs?

Diffusion model composition

K diffusion models: $s_{\theta^k}(x_t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(x_0)$ for $k = 1, \dots, K$

■ CONJUNCTION MODEL

$$p^{\text{conjunction}}(x_0) \approx \prod_{k=1}^K p^k(x_0)$$

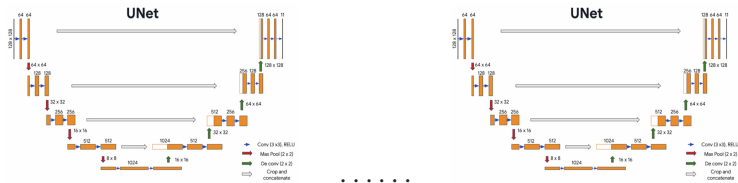
■ NEGATION MODEL

$$p^{\text{negation}}(x_0) \approx \frac{p^1(x_0)}{p^2(x_0)}$$

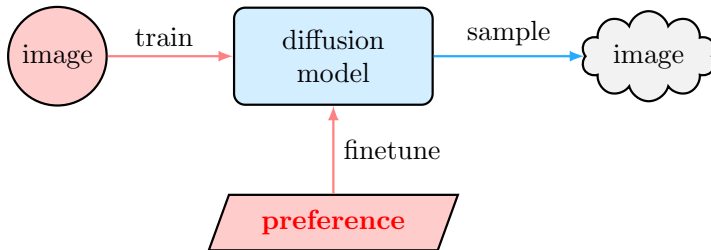
■ DISJUNCTION MODEL

$$p^{\text{disjunction}}(x_0) \approx \sum_{k=1}^K p^k(x_0)$$

Alignment + composition framework



$$x_0 \sim p^1(\cdot), \dots, p^K(\cdot)$$



preference – reward / human value

Outline

- RECAP
- REGULARIZATION-BASED APPROACH

Diffusion model composition: conjunction

K diffusion models: $s_{\theta^k}(x_t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(x_0)$ for $k = 1, \dots, K$

Reward model: $r(x_0)$ for a sample x_0

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(p(\cdot) \parallel p^k(\cdot))$$

$\beta_1, \dots, \beta_K$ – tradeoff parameters

★ optimal model $p^\star$

$$p^\star(x_0) \propto p^{\text{con}}(x_0) e^{r(x_0)/\mathbf{1}^\top \beta} \quad \text{and} \quad p^{\text{con}}(x_0) = \prod_{k=1}^K (p^k(x_0))^{\beta_k / \mathbf{1}^\top \beta} \times \frac{1}{Z}$$

Exponentially weighted conjunction distribution

■ KL DIVERGENCE SUMMATION

$$\begin{aligned}\sum_{k=1}^K \beta_k D_{\text{KL}}(p(\cdot) \parallel p^k(\cdot)) &= \sum_{k=1}^K \beta_k \mathbb{E}_{x_0 \sim p(\cdot)} \left[\log p(x_0) - \log p^k(x_0) \right] \\&= \mathbf{1}^\top \beta \mathbb{E}_{x_0 \sim p(\cdot)} [\log p(x_0)] \\&\quad - \mathbf{1}^\top \beta \mathbb{E}_{x_0 \sim p(\cdot)} \left[\log \prod_{k=1}^K (p^k(x_0))^{\beta_k / \mathbf{1}^\top \beta} \right] \\&= \mathbf{1}^\top \beta \mathbb{E}_{x_0 \sim p(\cdot)} [\log p(x_0)] \\&\quad - \mathbf{1}^\top \beta \mathbb{E}_{x_0 \sim p(\cdot)} [\log p^{\text{con}}(x_0)] - \mathbf{1}^\top \beta \log Z \\&= \mathbf{1}^\top \beta D_{\text{KL}}(p(\cdot) \parallel p^{\text{con}}(\cdot)) - \underbrace{\mathbf{1}^\top \beta \log Z}_{p\text{-independent}}\end{aligned}$$

KL divergence summation $\implies$ distribution conjunction

■ CONJUNCTION-BASED ALIGNMENT

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(p(\cdot) \| p^k(\cdot))$$

$$\Updownarrow$$

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \mathbf{1}^\top \beta D_{\text{KL}}(p(\cdot) \| p^{\text{con}}(\cdot))$$

★ conjunction distribution p^{con}

$$p^{\text{con}}(x_0) = \prod_{k=1}^K (p^k(x_0))^{\beta_k / \mathbf{1}^\top \beta} \times \frac{1}{Z}$$

Relaxation of point-wise KL divergence as path-wise one

Diffusion model composition: negation

K diffusion models: $s_{\theta^k}(x_t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(x_0)$ for $k = 1, \dots, K$

Reward model: $r(x_0)$ for a sample x_0

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \beta_1 D_{\text{KL}}(p(\cdot) \| p^1(\cdot)) + \beta_2 D_{\text{KL}}(p(\cdot) \| p^2(\cdot))$$

$\beta_1, \beta_2 \geq 0$, and $\beta_1 > \beta_2$ – tradeoff parameters

★ optimal model $p^\star$

$$p^\star(x_0) \propto p^{\text{neg}}(x_0) e^{r(x_0)/(\beta_1 - \beta_2)} \quad \text{and} \quad p^{\text{neg}}(x_0) = \frac{(p^1(x_0))^{\frac{\beta_1}{\beta_1 - \beta_2}}}{(p^2(x_0))^{\frac{\beta_2}{\beta_1 - \beta_2}}} \times \frac{1}{Z}$$

Exponentially weighted negation distribution

■ KL DIVERGENCE DIFFERENCE

$$\begin{aligned} & \beta_1 D_{\text{KL}}(p(\cdot) \parallel p^1(\cdot)) - \beta_2 D_{\text{KL}}(p(\cdot) \parallel p^2(\cdot)) \\ &= \beta_1 \mathbb{E}_{x_0 \sim p(\cdot)} [\log p(x_0) - \log p^1(x_0)] - \beta_2 \mathbb{E}_{x_0 \sim p(\cdot)} [\log p(x_0) - \log p^2(x_0)] \\ &= (\beta_1 - \beta_2) \mathbb{E}_{x_0 \sim p(\cdot)} [\log p(x_0)] - (\beta_1 - \beta_2) \mathbb{E}_{x_0 \sim p(\cdot)} \left[\log \frac{(p^1(x_0))^{\frac{\beta_1}{\beta_1 - \beta_2}}}{(p^2(x_0))^{\frac{\beta_2}{\beta_1 - \beta_2}}} \right] \\ &= (\beta_1 - \beta_2) \mathbb{E}_{x_0 \sim p(\cdot)} [\log p(x_0)] - (\beta_1 - \beta_2) \mathbb{E}_{x_0 \sim p(\cdot)} [\log p^{\text{neg}}(x_0)] \\ &\quad - (\beta_1 - \beta_2) \log Z \\ &= (\beta_1 - \beta_2) D_{\text{KL}}(p(\cdot) \parallel p^{\text{neg}}(\cdot)) - \underbrace{(\beta_1 - \beta_2) \log Z}_{p\text{-independent}} \end{aligned}$$

$$\beta_1 > \beta_2 \geq 0$$

KL divergence difference $\implies$ distribution negation

■ NEGATION-BASED ALIGNMENT

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \beta_1 D_{\text{KL}}(p(\cdot) \| p^1(\cdot)) + \beta_2 D_{\text{KL}}(p(\cdot) \| p^2(\cdot))$$

$$\Updownarrow$$

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - (\beta_1 - \beta_2) D_{\text{KL}}(p(\cdot) \| p^{\text{neg}}(\cdot))$$

★ negation distribution p^{pre}

$$p^{\text{neg}}(x_0) = \frac{(p^1(x_0))^{\frac{\beta_1}{\beta_1 - \beta_2}}}{(p^2(x_0))^{\frac{\beta_2}{\beta_1 - \beta_2}}} \times \frac{1}{Z}$$

$$\beta_1 > \beta_2$$

Relaxation of point-wise KL divergence as path-wise one

Diffusion model composition: disjunction

K diffusion models: $s_{\theta^k}(x_t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(x_0)$ for $k = 1, \dots, K$

Reward model: $r(x_0)$ for a sample x_0

■ DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(p^k(\cdot) \| p(\cdot))$$

$\beta_1, \dots, \beta_K$ – tradeoff parameters

★ optimal model $p^\star$

$$p^\star(x_0) \propto \frac{1}{r(x_0)} p^{\text{dis}}(x_0) \quad \text{and} \quad p^{\text{dis}}(x_0) = \sum_{k=1}^K \beta_k p^k(x_0)$$

Reciprocally weighted disjunction distribution

■ FORWARD KL DIVERGENCE SUMMATION

$$\begin{aligned}
 \sum_{k=1}^K \beta_k D_{\text{KL}}(p^k(\cdot) \parallel p(\cdot)) &= \sum_{k=1}^K \beta_k \mathbb{E}_{x_0 \sim p^k(\cdot)} [\log p^k(x_0) - \log p(x_0)] \\
 &= \sum_{k=1}^K \beta_k \mathbb{E}_{x_0 \sim p^k(\cdot)} [\log p^k(x_0)] \\
 &\quad - \mathbf{1}^\top \beta \mathbb{E}_{x_0 \sim \sum_{k=1}^K \frac{\beta_k}{\mathbf{1}^\top \beta} p^k(x_0)} [\log p(x_0)] \\
 &= \sum_{k=1}^K \beta_k \mathbb{E}_{x_0 \sim p^k(\cdot)} [\log p^k(x_0)] - \mathbf{1}^\top \beta \mathbb{E}_{x_0 \sim p^{\text{dis}}(\cdot)} [\log p^{\text{dis}}(x_0)] \\
 &\quad + \mathbf{1}^\top \beta \mathbb{E}_{x_0 \sim p^{\text{dis}}(\cdot)} [\log p^{\text{dis}}(x_0)] - \mathbf{1}^\top \beta \mathbb{E}_{x_0 \sim p^{\text{dis}}(\cdot)} [\log p(x_0)] \\
 &= \underbrace{- \sum_{k=1}^K \beta_k (H(p^{\text{dis}}) - H(p^k)) + \mathbf{1}^\top \beta D_{\text{KL}}(p^{\text{dis}}(\cdot) \parallel p(\cdot))}_{p\text{-independent}}
 \end{aligned}$$

Reverse KL divergence summation $\implies$ distribution disjunction

■ DISJUNCTION-BASED ALIGNMENT

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(p^k(\cdot) \parallel p(\cdot))$$

$$\Updownarrow$$

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \mathbf{1}^\top \beta D_{\text{KL}}(p^{\text{dis}}(\cdot) \parallel p(\cdot))$$

★ disjunction distribution p^{dis}

$$p^{\text{dis}}(x_0) = -\frac{1}{r(x_0)} \sum_{k=1}^K \beta_k p^k(x_0) \times \frac{1}{Z}$$

Relaxation of point-wise KL divergence as path-wise one

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Diffusion model composition: conjunction

K diffusion models: $s_{\theta^k}(x_t, t)$ for $k = 1, \dots, K$

K target distributions: $p^k(x_0)$ for $k = 1, \dots, K$

Reward model: $r(x_0)$ for a sample x_0

■ CONSTRAINED DISTRIBUTION OPTIMIZATION

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)]$$

$$\text{subject to} \quad D_{\text{KL}}(p(\cdot) \parallel p^k(\cdot)) \leq \epsilon_k$$

for $k = 1, \dots, K$

★ linear objective function and convex constraints

High-dimensional, convex $\implies$ strong duality

Lagrangian approach

■ LAGRANGIAN

$$L(p; \lambda) = \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \sum_{k=1}^K \lambda_k D_{\text{KL}}(p(\cdot) \parallel p^k(\cdot)) + \lambda^\top \epsilon$$

$$\lambda^\top \epsilon := \sum_{k=1}^K \lambda_k \epsilon_k$$

■ SINGLE-OBJECTIVE ALIGNMENT

$$\underset{p}{\text{maximize}} \quad L(p; \lambda)$$

★ explicit maximizer $p^*(\lambda)$

Unknown optimal dual variable λ^*

Strong duality

Dual problem: $\underset{\lambda}{\text{minimize}} \underset{p}{\text{maximize}} L(p; \lambda)$

Primal problem: $\underset{p}{\text{maximize}} \underset{\lambda}{\text{minimize}} L(p; \lambda)$

■ CONVEX-CONCAVE MIN-MAX OPTIMIZATION

$$\underset{\lambda}{\text{minimize}} \underset{p}{\text{maximize}} L(p; \lambda) = \underset{p}{\text{maximize}} \underset{\lambda}{\text{minimize}} L(p; \lambda)$$

Strong duality $\implies$ **exchange** of min and max

OBJECTIVE: find a minimax point (p^*, λ^*)

Dual problem

$$\underset{\lambda}{\text{minimize}} \quad \underset{p}{\text{maximize}} \quad L(p; \lambda)$$

$$L(p; \lambda) := \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \sum_{k=1}^K \lambda_k D_{\text{KL}}(p(\cdot) \| p^k(\cdot)) + \lambda^\top \epsilon$$

■ SINGLE-OBJECTIVE ALIGNMENT

$$D(\lambda) := \underset{p}{\text{maximize}} \quad L(p; \lambda)$$

★ explicit maximizer $p^*(x_0; \lambda)$

$$p^*(x_0; \lambda) \propto p^{\text{con}}(x_0; \lambda) e^{r(x_0)/\mathbf{1}^\top \lambda} \quad \text{and} \quad p^{\text{con}}(x_0; \lambda) = \prod_{k=1}^K (p^k(x_0))^{\lambda_k / \mathbf{1}^\top \lambda} \times \frac{1}{Z}$$

Optimal dualization

■ EXPLICIT DUAL FUNCTION

$$D(\lambda) = \mathbf{1}^\top \lambda \log \mathbb{E}_{x_0 \sim p^{\text{con}}(\cdot; \lambda)} \left[\exp \left(\frac{r(x_0) + \lambda^\top \epsilon}{\mathbf{1}^\top \lambda} \right) \right] - \mathbf{1}^\top \lambda \log Z$$

$$p^{\text{con}}(x_0; \lambda) = \prod_{k=1}^K (p^k(x_0))^{\lambda_k / \mathbf{1}^\top \lambda} \times \frac{1}{Z}$$

Independent of p , and **convexity**

$\implies ??$ **easy to find** λ^*

■ SAMPLES FROM COMPOSED MODEL

$$s_{\theta^{\text{soup}}}(x_t, t) = s_{\sum_{k=1}^K \frac{\lambda_k}{\mathbf{1}^\top \lambda}} \theta^k(x_t, t)$$

Soup model

■ NORMALIZATION CONSTANT

$$\mathbf{1}^\top \lambda \log Z \approx \prod_{k=1}^K (p^k(x'_0))^{\lambda_k}$$

Max-approximation requires density estimation $\implies$ Intractable

Primal-dual algorithm

■ PRIMAL UPDATE

$$p^{(t)}(\cdot; \lambda^{(t)}) \leftarrow \operatorname{argmax}_p L(p; \lambda^{(t)})$$

$$p^{(t)}(x_0; \lambda) \propto p^{\text{con}}(x_0; \lambda^{(t)}) e^{r(x_0)/\mathbf{1}^\top \lambda^{(t)}}$$

■ DUAL UPDATE

$$\lambda_k^{(t+1)} \leftarrow \mathcal{P}_{\geq 0} \left(\lambda_k^{(t)} - \eta \left(\epsilon_k - D_{\text{KL}}(p^{(t)}(\cdot; \lambda^{(t)}) \| p^k(\cdot)) \right) \right)$$

Dual subgradient descent

$\Rightarrow$ guaranteed to find λ^* , **while slow convergence**
and high computational cost

Summary

■ RECAP

■ REGULARIZATION-BASED APPROACH

KL divergence summation $\equiv$ conjunction, KL divergence difference $\equiv$ negation,
forward KL divergence summation $\equiv$ disjunction

■ CONSTRAINED APPROACH

KL divergence constraints for conjunction (similarly, negation and disjunction),
primal-dual algorithm